systems TECH



PRODUCT RANGE

Digital Tachometers, Speed Sensors and Wireless Sensors



Handheld and panel mount tachometers, tacho-generators, magnetic pick-up, proximity and photo-reflective sensors, temperature and humidity sensors.





Authorised distributor in India

CIE Instruments

Cambridge Instruments & Engg. Co.

www.cieinstruments.in

Models at a glance

Handheld digital tachometers

HTM 590	Contact Type Digital Tachometer	p. 4
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HTM 890	Photo Contact Digital Tachometer	p. 5

Digital panel mount tachometers

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TM-802	Digital Panel Mount Tachometer with Magnetic Pick-up	p. 8
TM-803	Digital Panel Mount Tachometer with Proximity Switch	p. 10
TM-804	Digital Panel Mount Tachometer with Photo-Reflective Sensor	p. 12

Tachometer speed sensors

TM-801 TG	Digital Tachogenerator	p. 7
TM-802 MPU	Magnetic Pick-up Sensor	p. 9
TM-803 PS	Proximity Switch Sensor	p. 11
TM-804 PR	Photo-Reflective Sensor	p. 13
TM-802 UNF	5/8 UNF Magnetic Pick-up Sensor	

Wireless temperature and humidity sensors

TS-200	Wireless Temperature Sensor	p. 14
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Models without a datasheet page: TM-802 UNF pick-up (see cieinstruments.in).

HTM-560 NON-CONTACT TYPE DIGITAL TACHOMETER

MODEL NUMBER - HTM 560

KEY FEATURES TRUSTED FOR QUALITY AND RELIABILITY

- Full Digital, Micro Controller LSI circuit design for high reliability and versatility.
- Comes with a sturdy, easy to carry/handle carrying case box with foam padding for protection against impact.
- Ergonomically designed for handheld based operations for risk free RPM measurement at all times.
- Excellent Quality, compact size and accurate reading.
- Economic and effective Battery power utilisation.



TECHNICAL SPECIFICATIONS

Ranges:	60 RPM – 1,00,000 RPM
Resolution:	0.1 upto 1000 RPM/1 above 1000 RPM 0.01 upto 100 meter/min/0.1 above 100 meter/min
Display Type:	5 digits, 10mm(0.38") height RED LED
Accuracy:	+/- 0.05 + 1 Digit
Sampling Rate:	1 second
Time Base:	Quartz Crystal Controlled
Measuring Distance:	75mm (3 inch) to 300mm (12 inch)
Low Battery:	Indicated by "Lobat" when the battery voltage drops below operating voltage
Memory Recall:	'Last', 'Max', 'Min' Values memorized automatically, upto 120sec can be recalled
Power Source:	4 X 1.5V AA Battery. Auto Shut off feature included
Battery Life:	90hrs for alkaline batteries | 30hrs for Carbon-Zinc batteries
Operating Temp:	0° Celsius to 60° Celsius (32° F to 140° F)
Size:	70 x 200 x 40mm (2.8 x 7.8 x 1.6 inch)
Weight:	270grams including battery
Standard Accessories:	Reflective tape (Marks) 1000x12mm(40×0.5 inch) 1 Padded Carrying Case 1

HTM-590 CONTACT TYPE DIGITAL TACHOMETER

MODEL NUMBER - HTM 590



KEY FEATURES TRUSTED FOR QUALITY AND RELIABILITY

- Full Digital, Micro Controller LSI circuit design for high reliability and versatility.
- Comes with a sturdy, easy to carry/handle carrying case box with foam padding for protection against impact.
- Ergonomically designed for handheld based operations for risk free RPM measurement at all times.
- Excellent Quality, compact size and accurate reading.
- Economic and effective Battery power utilisation.

TECHNICAL SPECIFICATIONS

Ranges:	60 RPM – 50,000 RPM 6m/min – 5000 m/min (Surface speed)
Resolution:	0.1 upto 1000 RPM/1 above 1000 RPM 0.01 upto 100 meter/min/0.1 above 100 meter/min
Display Type:	5 digits, 10mm(0.38") height RED LED
Accuracy:	+/- 0.05 + 1 Digit
Sampling Rate:	1 second
Time Base:	Quartz Crystal Controlled
Low Battery:	Indicated by "Lobat" when the battery voltage drops below operating voltage
Memory Recall:	'Last', 'Max', 'Min' Values memorized automatically, upto 120sec can be recalled
Power Source:	4 X 1.5V AA Battery. Auto Shut off feature included
Battery Life:	90hrs for alkaline batteries | 30hrs for Carbon-Zinc batteries
Operating Temp:	0° Celsius to 60° Celsius (32° F to 140° F)
Size:	70 x 200 x 40mm (2.8 x 7.8 x 1.6 inch)
Weight:	320 grams

Standard Accessories:		
	Male rubber contact tip	2
	Female rubber contact tip	2
	Peripheral rubber contact tip reflective tape (Marks)	1
	Padded Carrying Case	1
	User manual	1

HTM-890 CONTACT TYPE DIGITAL TACHOMETER

MODEL NUMBER – HTM 890



KEY FEATURES USED IN INDUSTRIAL APPLICATIONS TO MONITOR THE PERFORMANCE OF MOTORS

- Full Digital, Micro Controller LSI circuit design for high reliability and versatility.
- Sturdy carrying case with foam padding for impact protection
- Ergonomically designed for handheld based operations for risk free RPM measurement at all times.
- Excellent Quality, compact size and accurate reading.
- Economic and effective Battery power utilisation.



TECHNICAL SPECIFICATIONS

Ranges:	60 RPM – 50,000 RPM (contact side) 6 m/min – 5000 m/min (surface speed) 60 RPM – 1,00,000 RPM (photo side)
Resolution:	0.1 RPM: 0 – 1000 RPM | 1 RPM: > 1000 RPM 0.01 : 0 – 100 meter/min | 0.1 : > 100 meter/min
Display Type:	5 digits, 10mm(0.38") height RED LED
Accuracy:	+/- 0.05 + 1 Digit
Sampling Rate:	1 second
Time Base:	Quartz Crystal Controlled
Low Battery:	Indicated by "Lobat" when the battery voltage drops below operating voltage
Memory Recall:	'Last', 'Max', 'Min' Values memorized automatically, upto 120sec can be recalled
Power Source:	4 X 1.5V AA Battery. Auto Shut off feature included
Battery Life:	90hrs for alkaline batteries (contact side) 30hrs for Carbon-Zinc batteries (contact side) 24hrs for alkaline batteries (Photo side) 12hrs for Carbon-Zinc batteries (Photo side)
Operating Temp:	0° Celsius to 60° Celsius (32° F to 140° F)
Size:	70 x 200 x 40mm (2.8 x 7.8 x 1.6 inch)
Weight:	320grams including battery
Standard Accessories:	Male rubber contact tip 2 Female rubber contact tip 2 Peripheral rubber contact tip reflective tape (Marks) 1000x12mm(40×0.5 inch) 1 Padded Carrying Case 1 User manual 1

DIGITAL PANEL MOUNT TACHOMETER WITH DIGITAL TACHOGENERATOR SENSOR

MODEL NUMBER – TM 801



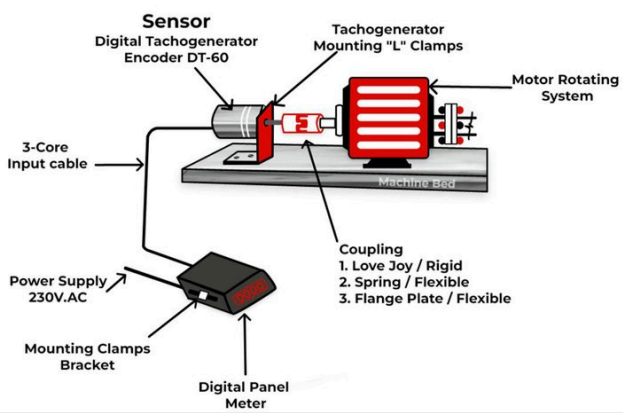
KEY FEATURES

- Used in industrial applications to monitor the performance of motors.
- Full Digital, Micro Controller LSI circuit design with Auto Ranging, Crystal controlled time base.
- Useful for measurements of RPMs of motors, mixtures, pumps and other rotary devices. With excellent quality, compact size and accurate reading.

TECHNICAL SPECIFICATIONS

Ranges:	60 RPM – 50,000 RPM
Resolution:	0.1 RPM from 60 to 1000 RPM 1 RPM from 1000 to 50,000 RPM
Display Type:	5 digits, 12.5mm(0.5") height RED LED
Accuracy:	+/- 0.05 + 1 Least count
Sampling Rate:	1 Sample per second
Time Base:	Quartz Crystal Controlled
Power Source:	220 Volts A.C. 50Hz +/- 10%
Panel Size:	96mm x 48mm x 130mm (96mm Sq)
Standard Accessories:	Tachogenerator sensor 1 3.0 Meter Shielded cable with connectors 1 User manual 1





DIGITAL TACHOGENERATOR

MODEL NUMBER – TM 801



SPECIFICATIONS: ELECTRONIC

Supply Voltage:	6V DC (Regulated)
Supply Current:	50mamp – 70mamp
Output Type:	Digital Pulse Output
O/P Pulse Rate:	60ppr (Pulses per Revolutions)
O/P Pulse Amplitude:	Same as Supply Voltage.
O/P Frequency:	10 KHz @ 10000 RPM.
O/P Characteristics:	Maximum ripple over 1revolution = 15% Electrical Phase Error per increment = +20%
Outlet Cable:	2/3 meter long (Standard) with three cores along with connector.

SPECIFICATIONS: MECHANICAL

Electronic Body:	Machined Aluminium Housing with Aluminium or ABS plastic cover.
Shaft Size:	10.0 mm (Dia.), 23 mm shaft extension.. Stainless Steel (SS-304).
Mounting:	Servo or Face – by 3 screws of M-4 (1200 apart on a PCD of 48 mm dia.).
Overall Size:	Body dia. 62 mm, Depth: 80 mm. (excluding shaft and connector gland).
Weight(Approx.):	300 gram.

DIGITAL PANEL MOUNT TACHOMETER WITH MAGNETIC PICK-UP SENSOR

MODEL NUMBER – TM 802

KEY FEATURES

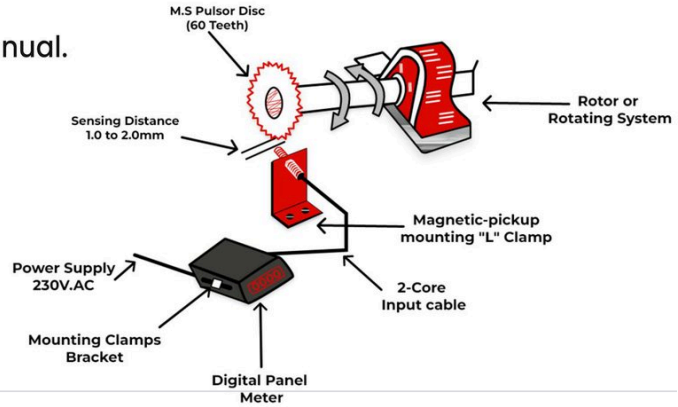


- Used in industrial applications to monitor the performance of motors.
- Full Digital, Micro Controller LSI circuit design for high reliability and versatility
- Comes with a sturdy, easy to carry/handle carrying case box with foam padding for protection against impact.

TECHNICAL SPECIFICATIONS

Ranges:	60 RPM – 50,000 RPM
Resolution:	0.1 RPM from 60 to 1000 RPM 1 RPM from 1000 to 50,000 RPM
Display Type:	5 digits, 12.5mm(0.5") height RED LED
Accuracy:	+/- 0.05 + 1 Least count
Sampling Rate:	1 Sample per second
Time Base:	Quartz Crystal Controlled
Power Source:	220 Volts A.C. 50Hz +/- 10%
Panel Size:	96mm x 48mm x 130mm (96mm Sq)
Standard Accessories:	Magnetic pick-up sensor with 3.0 Meter Shielded cable with connectors and instruction manual.





TM 802 MAGNETIC PICKUP SENSOR

MODEL NUMBER – TM 802

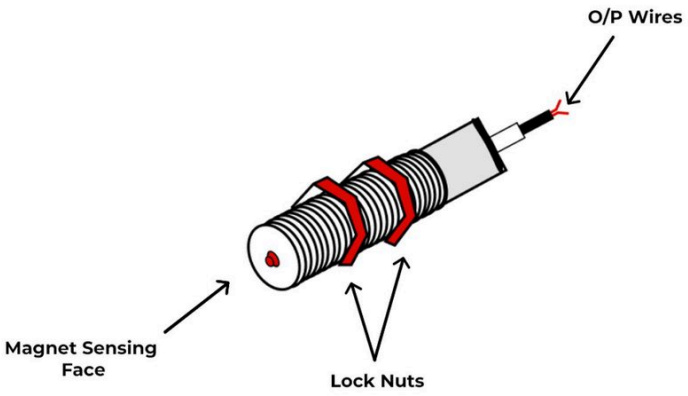


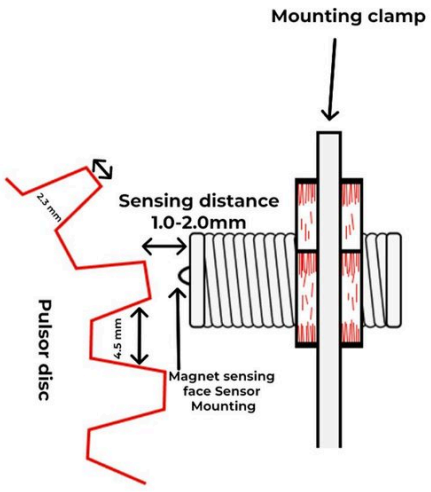
SPECIFICATIONS: ELECTRONIC

Supply Voltage:	6V DC (Regulated).
Supply Current:	50 mamp – 70 amp
Output Type:	Digital Pulse Output
O/P Pulse Rate:	60ppr (Pulses per Revolutions)
O/P Pulse Amplitude:	Same as Supply Voltage.
O/P Frequency:	10 KHz @ 10000 RPM.
Outlet Cable:	2/3 meter long (Standard) with three cores .

SPECIFICATIONS: MECHANICAL

Mechanical Body:	Machined BRASS/SS threaded housing with two locknuts.
Mounting:	Face – by 2 locknuts of M16
Overall Size:	Body dia – 18 mm , Length : 80 mm.
Weight(Approx.):	100 gram.





DIGITAL PANEL MOUNT TACHOMETER WITH PROXIMITY SWITCH SENSOR

MODEL NUMBER – TM 803

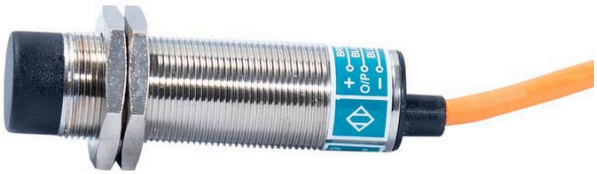
KEY FEATURES

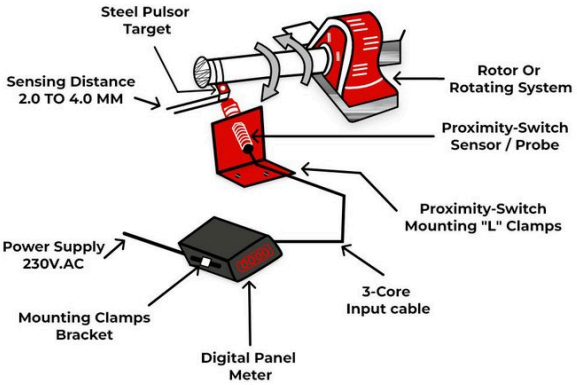
- Used in industrial applications to monitor the performance of motors.
- Full Digital, Micro Controller LSI circuit design for high reliability and versatility
- Comes with a sturdy, easy to carry/handle carrying case box with foam padding for protection against impact.



TECHNICAL SPECIFICATIONS

Ranges:	60 RPM – 50,000 RPM
Resolution:	0.1 RPM from 60 to 1000 RPM 1 RPM from 1000 to 50,000 RPM
Display Type:	5 digits, 12.5mm(0.5") height RED LED
Accuracy:	+/- 0.05 + 1 Least count
Sampling Rate:	1 Sample per second
Time Base:	Quartz Crystal Controlled
Power Source:	220 Volts A.C. 50Hz +/- 10%
Panel Size:	96mm x 48mm x 130mm (96mm Sq)
Standard Accessories:	Proximity Switch sensor with 3.0 Meter Shielded cable with connectors and instruction manual.





PROXIMITY SWITCH SENSOR

MODEL NUMBER – TM 803

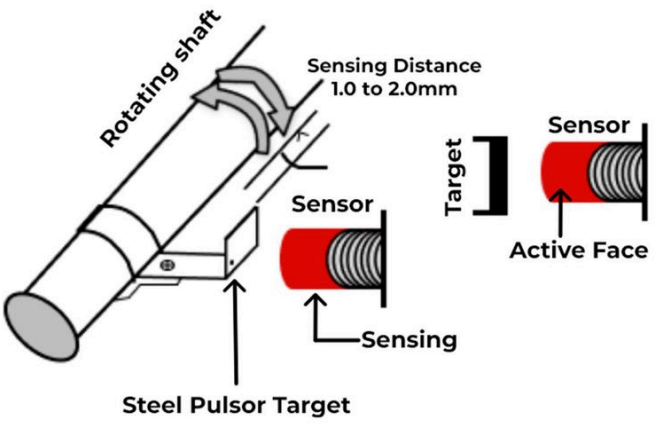
SPECIFICATIONS: ELECTRONIC

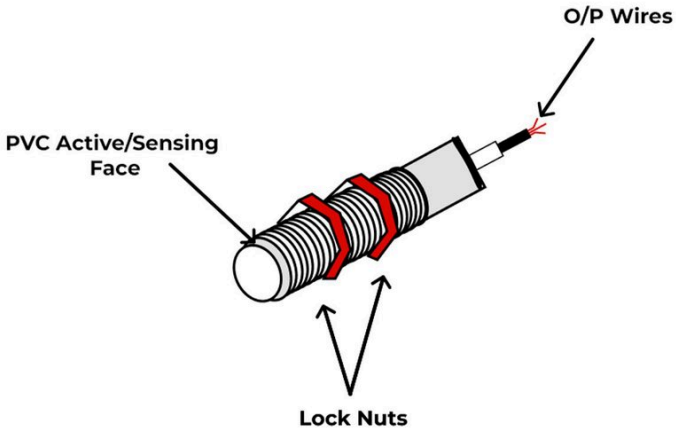


Supply Voltage:	5V to 24V DC (Regulated)
Supply Current:	50 mA – 70 mA
Output Type:	Digital Pulse Output: 3-wire
O/P Pulse Rate:	1 ppr (Pulses per Revolutions)
O/P Pulse Amplitude:	Same as Supply Voltage.
Signal Indicator:	10 KHz @ 10000 RPM.
O/P Characteristics:	Green LED provided at back
Outlet Cable:	2/3 meter long (Standard) with three cores .

SPECIFICATIONS: MECHANICAL

Mechanical Body:	Machined BRASS/SS threaded housing with two locknuts.
Mounting:	Face – by 2 locknuts of M18
Overall Size:	Body dia – 18 mm, Length: 65mm.
Weight(Approx.):	100 gram.





DIGITAL PANEL MOUNT TACHOMETER WITH PHOTO-REFLECTIVE SENSOR

MODEL NUMBER – TM 804

KEY FEATURES

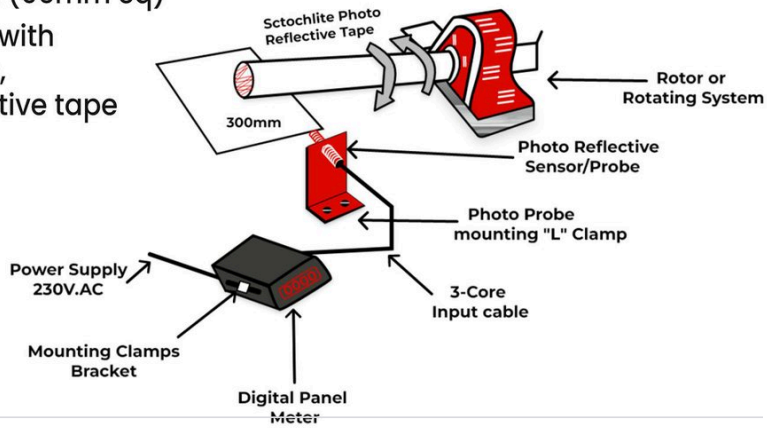


- Used in industrial applications to monitor the performance of motors.
- Full Digital, Micro Controller LSI circuit design with Auto Ranging, Crystal controlled time base.
- Useful for measurements of RPMs of motors, mixtures, pumps and other rotary devices. With excellent quality, compact size and accurate reading.

TECHNICAL SPECIFICATIONS

Ranges:	60 RPM – 50,000 RPM
Resolution:	0.1 RPM from 60 to 1000 RPM 1 RPM from 1000 to 50,000 RPM
Display Type:	5 digits, 12.5mm(0.5") height RED LED
Accuracy:	+/- 0.05 + 1 Least count
Sampling Rate:	1 Sample per second
Time Base:	Quartz Crystal Controlled
Power Source:	220 Volts A.C. 50Hz +/- 10%
Panel Size:	96mm x 48mm x 130mm (96mm Sq)
Standard Accessories:	Photo-Reflective sensor with 3.0 Meter Shielded cable, 12x1000 mm Photo reflective tape and instruction manual.





TM 804 PHOTO-REFLECTIVE SENSOR

MODEL NUMBER – TM 804

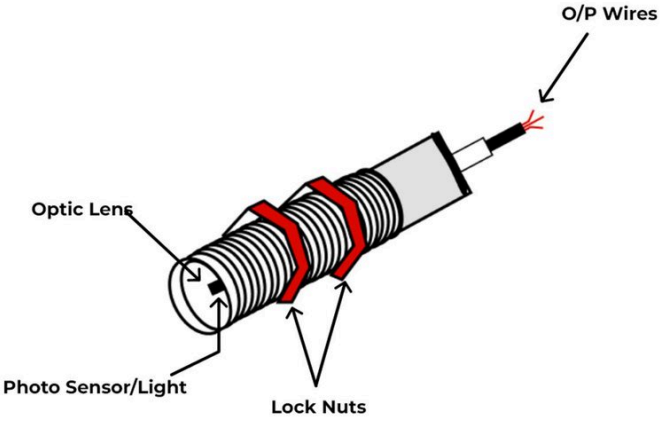
SPECIFICATIONS: ELECTRONIC



Supply Voltage:	6V DC (Regulated).
Supply Current:	125 mamp
Output Type:	Digital Pulse Output - 3 Wire
O/P Pulse Rate:	1ppr (Pulses per Revolutions)
O/P Pulse Amplitude:	+50 mV to 5 V.
O/P Frequency:	20 KHz @ 10000 RPM.
Outlet Cable:	2/3 meter long (Standard) with three cores along with connector.

SPECIFICATIONS: MECHANICAL

Mechanical Body:	Machined BRASS/SS threaded housing with two locknuts.
Mounting:	Face – by 2 locknuts of M22
Overall Size:	Body dia – 22 mm , Length : 75 mm.
Weight(Approx.):	150 gram.





TS-200 TEMPERATURE SENSOR

MODEL NUMBER - TS-200



KEY FEATURES

- Truly wireless –WiFi enabled (LoRaWAN version also available)
- Comes with a double sided sticker to easily mount it on the outer side of your refrigeration unit.
- Alarm temperature max, min (30, -20)
- Real-time, instant updates Readily available APIs for integration under 30 minutes!
- Temperature measurement range -50°C to 125°C Maximum resolution +/- 0.25°C
- Power Supply external microUSB. 5V, 500mA.

TECHNICAL SPECIFICATIONS

Sensor Ranges:	NDP –40°C to 125°C | DP: –55°C to +125°C
Accuracy:	NDP: ±0.5°C from –25°C to 85°C DP: ±0.5°C from -10°C to +85°C
Resolution:	0.0625°C
Power requirements:	5V DC micro USB max 30mA current draw (USB version)/ max 500mA current draw 2200mAh rechargeable battery (for Li-Ion version only) 2x AA Alkaline 1.5V cells (for non-rechargeable versions only)
Sensor cable:	50 cm, 1mm height, Flat Ribbon Cable
Processor:	32-bit processor running at 80Mhz and with a complete 2.4Ghz WiFi stack
Configurable parameters:	Measurement interval 60s Temperature delta to trigger data transmission +/- 1°C Alarm temperature max, min (30, -20)
Alarm:	Audio 70dB (min.) sweep tone Flashing screen with Alarm icon
Input:	Capacitive, non-mechanical touch sense x 1 (USB version only) Serial over USB commands (USB version only)

TEMPERATURE AND HUMIDITY 2IN1 SENSOR

MODEL NUMBER - TS-201



KEY FEATURES

TRUSTED QUALITY AND RELIABILITY

- Simple plug-and-use wireless device.
  Set up in 2 minutes and track worldwide.
- Auto log and view temperature and humidity of
  cold storage drugs and vaccines across multiple locations.
- Track real-time temperature+humidity and
  set thresholds for on-device audiovisual alarms.
- Designed to be used with existing refrigerators/freezers and cold rooms.
- Whatsapp and email alerts on temperature and humidity breaches.
  Downloadable and automated reports.
- Android and iOS mobile applications to manage alerts and access the dashboard.

TECHNICAL SPECIFICATIONS

Working Ranges:	-40°C to +125°C; 0% to 100%RH
Accuracy:	+/- 0.2°C and +/-2%RH
Long term drift:	<0.02°C/y & <0.25RH/y
Display Crisp:	OLED display 80dB alarm
Power supply:	5V DC micro USB power supply
Battery Backup:	Optional 8 hour battery back-up
Data Storage:	200 samples on-device data storage.